

- 4 A block of mass m oscillates vertically on a spring, as shown in Fig. 4.1.

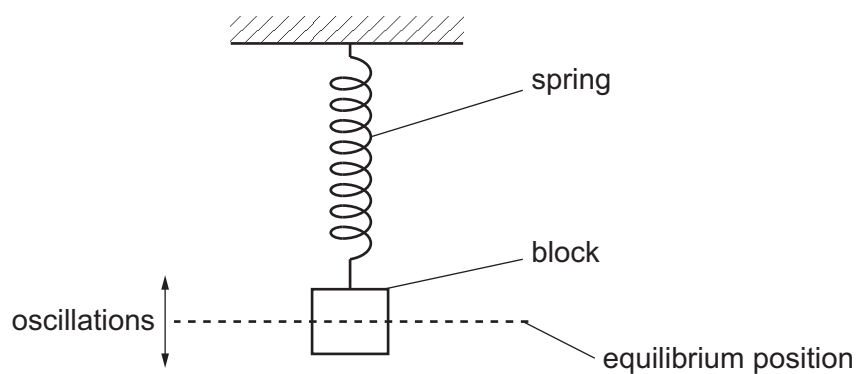


Fig. 4.1

The acceleration a of the block varies with displacement x from its equilibrium position, as shown in Fig. 4.2.

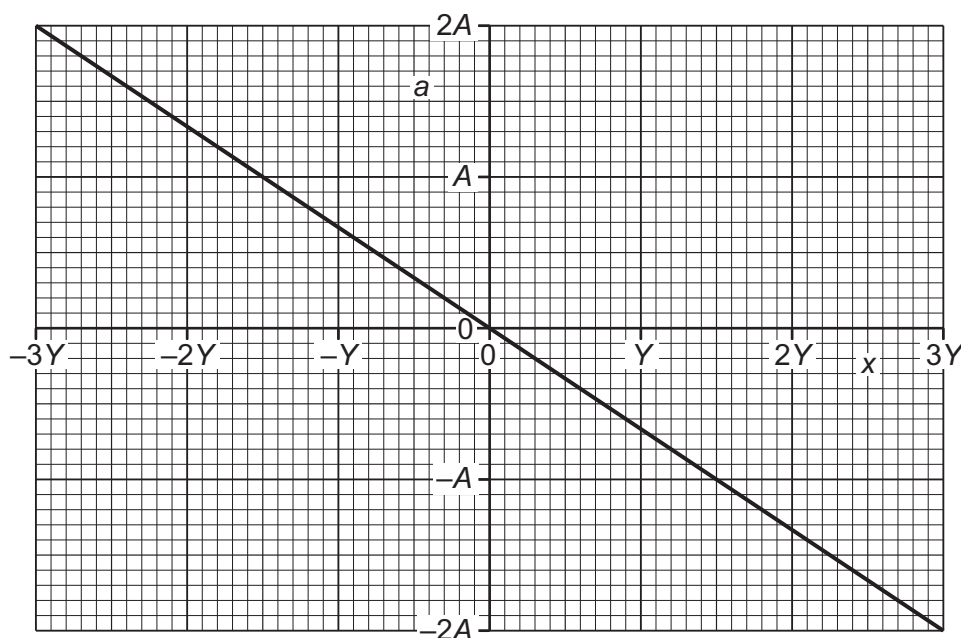


Fig. 4.2

The amplitude of the oscillations is $3Y$ and the maximum acceleration is $2A$.

- (a) Explain how Fig. 4.2 shows that the oscillations of the block are simple harmonic.

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.....

..... [2]



(b) Deduce expressions, in terms of some or all of m , A and Y , for:

(i) the angular frequency ω of the oscillations

$$\omega = \dots\dots\dots [1]$$

(ii) the maximum speed v_0 of the oscillations

$$v_0 = \dots\dots\dots [2]$$

(iii) the energy E of the oscillations.

$$E = \dots\dots\dots [2]$$

(c) The period of the oscillations is 0.75 s and the value of $3Y$ is 1.8 cm.

Determine an expression for x in terms of time t , where x is in cm and t is in seconds.

$$x = \dots\dots\dots [2]$$

[Total: 9]

